

YOINK

ORQ.AI

GREENPT

AIOPS COMMUNITY

Sustainable AI Hackathon

// build more with less AI

14:00 – 14:30 | Welcome & Networking

14:30 – 15:00 | Opening & Inspiration

15:00 – 18:30 | Hackathon

18:30 – 19:00 | Pizza

19:00 – 20:00 | Presentations & Awards



Welcome to the Sustainable AI Hackathon

- 14:00 – 14:30 Welcome & Networking
- 14:30 – 15:00 Opening & Inspiration
- 15:00 – 18:30 Hackathon
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Opening & Inspiration

- Introducing the Sponsors
- AI & Sustainability – A Vision for the Future
- Rules & Judging Criteria
- The Briefing



The all-in-one platform to build reliable LLM apps

Orq.ai's Generative AI Collaboration Platform is the #1 tool for engineering and product teams to build and ship LLM apps together.

A better world through efficient AI.

Everything you need from AI: chat, search, documents, API. Hosted in Europe.

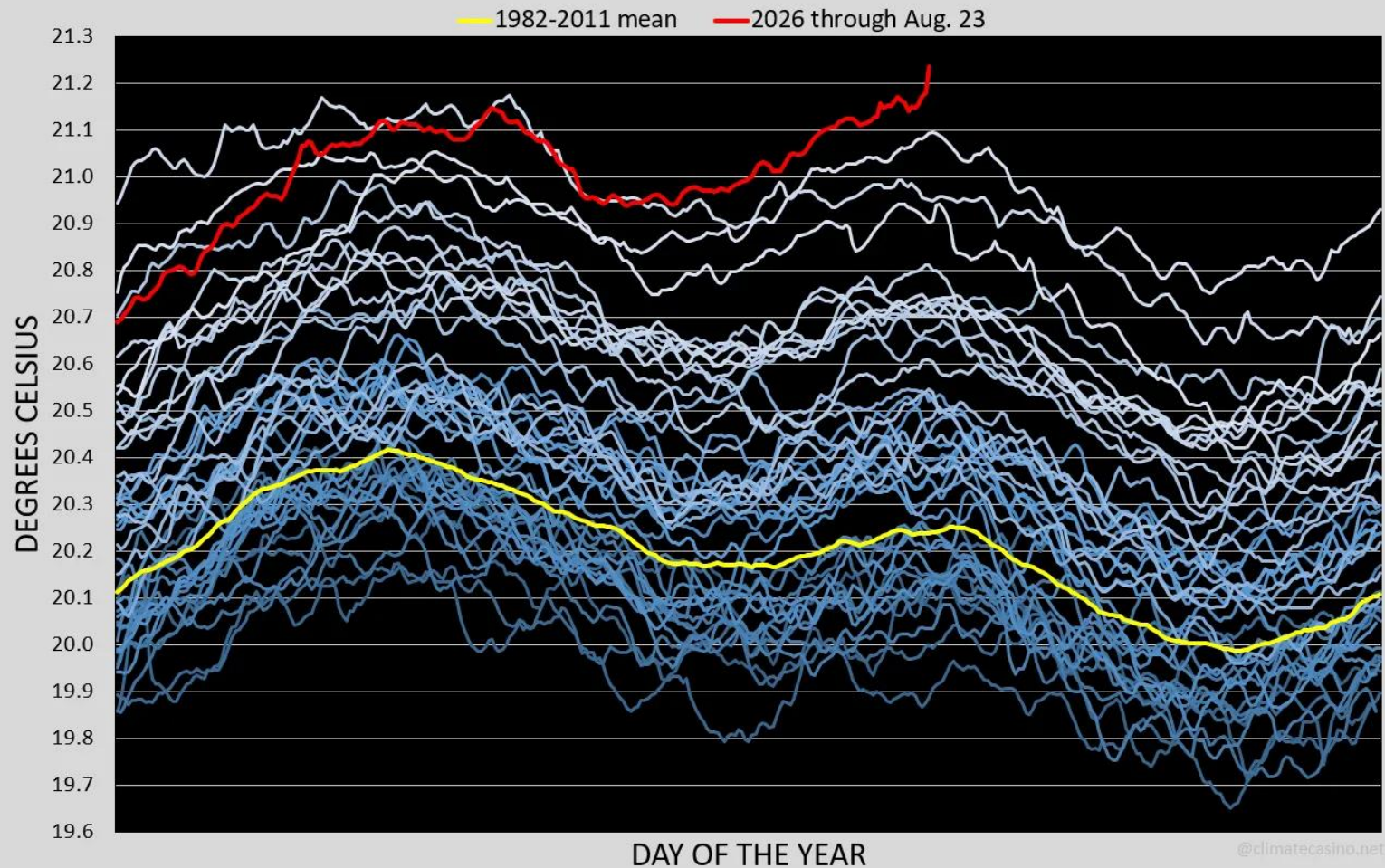
Powered by renewable energy. Built without compromises.

AI & Sustainability – A Vision for the Future



Global Sea-Surface Temperatures (60°S-60°N): 1982-2026

Data: https://climaterianalyzer.org/clim/sst_daily/json/oisst2.1_world2_sst_day.json



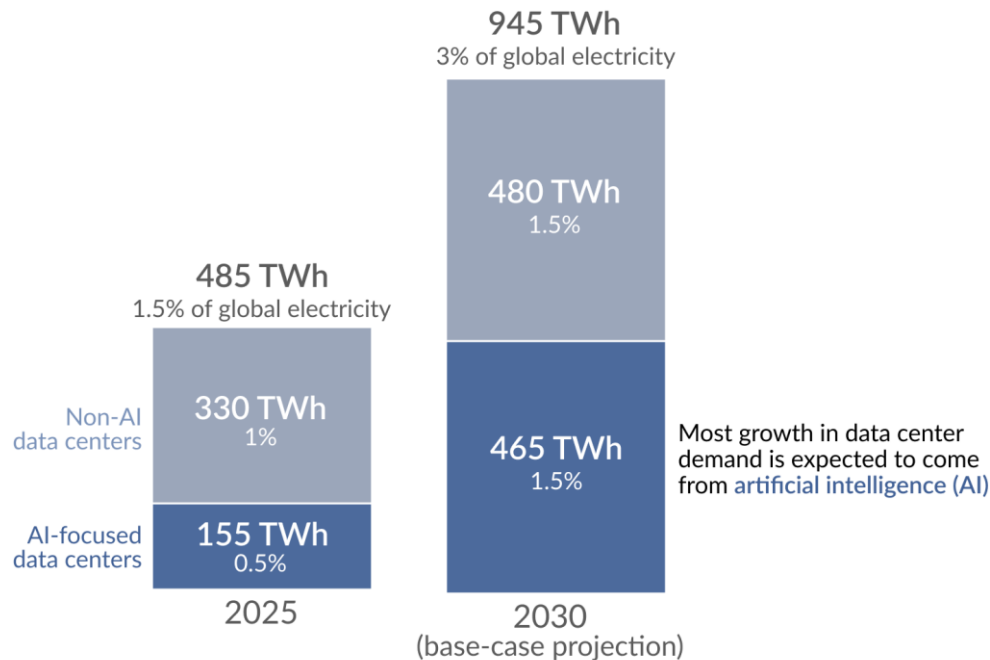
Global 2m Surface Temperatures: 2024 v. 2026

Data: Copernicus ERA5



How much of global electricity is used for data centers?

Measured in terawatt-hours (TWh). Estimates for 2025; base-case projections for 2030 from the International Energy Agency (IEA).



Note: Projections for 2030 are very uncertain, but shown here to reflect expectations that most data center growth will come from AI-focused ones.

Data source: IEA (2026). Key Questions on Energy and AI.

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Weer protest tegen datacenter, maar 'we weten allemaal dat we ze nodig hebben'

Tech bros can't agree on why everyone hates data centers

By [Truman Dickerson](#)

Protesters haul a guillotine to city council meeting about a potential AI data center, company rep cornered by protestors — 'it no longer felt safe to stay,' developer escorted out by police

News By [Jowi Morales](#) | Published August 7, 2026

Off with their heads!

What is “wrong” with AI?

- It is an arms race stimulating an infrastructure build-out
- It's explosive growth drives up the immediate need for extra electricity
- Renewables cannot scale fast enough to catch up
- The new automation-hype is making people anxious about the future
- ... and yet everyone is using it for everything



The AI Reality

- AI is here to stay, even if the current bubble pops
- “The genie is out of the bottle”
- “You can’t put the toothpaste back into the tube”
- ... pick your favourite 😂



Sustainable AI vs Green AI

- Green AI is the practice of optimising efficiency of AI systems
- Sustainable AI is the practice of using AI only where necessary



Sustainable AI

- Could a simpler solution solve the problem at lower energy?
 - AI is not the sustainable choice
- Does the AI solution replace a more energy-intensive process?
 - AI is a sustainable choice



Green AI

- Could a smaller model solve the problem?
- Can parts of the process be done by regular code / tools?
- Can we find an infrastructure provider with better emission data?
- Can we optimize prompts or workflows?

This is the goal of today's hackathon



The Future of Software Development

- What will our profession look like in 10 years?

10 Years ago..

Basically the same, with
different tools and frameworks

Now...

Honestly hard to say.



The Future of Software Development

AI “flops”

No more AI, back to how everything was before.

AI takes over

No more devs...



My Personal Take

- “Typing was never the problem”
- There’s a real difference between juniors+AI and seniors+AI
- The Vibe-coding works... until it doesn’t



Important Skills in the Age of AI

- Talking to non-technical people
- Understand requirements
- Translate requirements to technical decisions
- Document both requirements and technical decisions
- Understand how business and technology enhance each other

Wait a second... that's just software engineering!



My Vision for the Sustainable Future

- Most problems we face are fundamentally energy problems
- "Given infinite energy we could ..."
- As long as infinite (free) energy does not exist, sustainability practices are needed
- Every bit of energy we save, humanity as a whole can use for something else
- Hopefully to fix climate change, cure cancer, eliminate hunger, etc...



Let's take a small step in the right direction!



Rules & Judging Criteria



The Rules

- You can choose any stack
- You can choose any framework
- You can use any AI in preparation or brainstorming
- You must use GreenPT for the actual run
- You may not delegate to other AIs during a run
- You must start a run from an empty project



The Rules

- You may create skills
- You may use any AI tool, or create your own



Scoring Criteria – First Prize

- The application needs to be feature complete
- The application needs to be able to run
- The application needs to show the correct results
- Tie breaker criteria:
 - Code quality and maintainability
 - UX clarity, especially in the comparison view
- Extra criterium: you must be able to explain your solution 😄



Scoring Criteria – Creativity Prize

- The application does not need to be feature complete
- The application does not need to be able to run
- The application does not need to show the correct results
- The approach must be novel, innovative and promising



Configuring Your Tools

- Any OpenAI compatible tool should work
- The base URL is : `https://my.orq.ai/v3/router`
- The two available model IDs are:
 - `yoink@openailike/deepseek-v4-flash-0731`
 - `yoink@openailike/glm-5.2-caveman`
- Each team gets two API keys
- Use header `X-ORQ-IDENTITY-ID: team-X-run-Y` to differentiate



OpenCode example config

```
{
  "$schema": "https://opencode.ai/config.json",
  "provider": {
    "orq-yoink": {
      "npm": "@ai-sdk/openai-compatible",
      "name": "Orq.AI - Yoink Hackathon",
      "options": {
        "baseUrl": "https://my.orq.ai/v3/router"
      },
    },
    "models": {
      "yoink@openailike/yoink@openailike/deepseek-v4-flash-0731": {
        "name": "DeepSeek v4 Flash"
      },
      "yoink@openailike/yoink@openailike/glm-5.2-caveman": {
        "name": "GLM 5.2 - Caveman"
      }
    }
  }
}
```


Team Budgets

- Each team has a \$30 budget
- The models have different costs / 1M tokens:
 - DeepSeek : input \$0.14, output \$0.35
 - GLM-5.2 : input \$1.10, output \$4.40



Prizes – For both winners

- Turing Machine board game
- Six months GreenPT Pro
- Six months Orq.AI



The Briefing



The Application to Build

ISO 14040 – Lifecycle Management

Compares product lifecycles based on climate impact

Each product has a process, built from stages, built from flows

Each flow uses a material

Each material has emission factors



Impact Categories

ISO 14040 – Lifecycle Management

Category	Unit	What it measures
Global Warming Potential (GWP)	kg CO ₂ equivalent	Climate change contribution
Freshwater Eutrophication	kg PO ₄ equivalent	Nutrient pollution of waterways
Water Consumption	litres	Total freshwater consumed



Product Comparison

Normalised to: 1 litre of craft ale at point of consumer purchase

Craft amber ale, 330 ml glass bottle

1 litre of craft ale at point of consumer purchase

Scaling factor: 3.0303

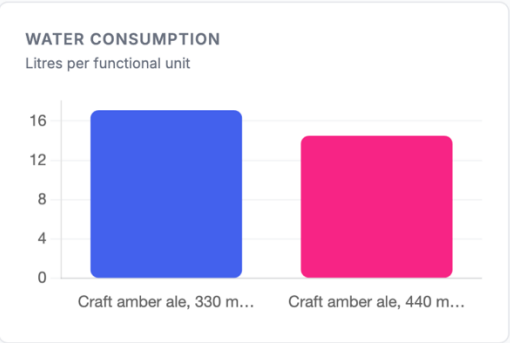
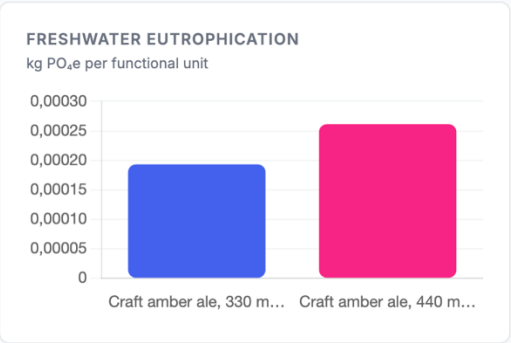
Craft amber ale, 440 ml tinplate steel can

1 litre of craft ale at point of consumer purchase

Scaling factor: 2.2727

Impact Category	Craft Amber Ale, 330 ml Glass Bottle	Craft Amber Ale, 440 ml Tinplate Steel Can	Unit	Verdict
Global Warming Potential GWP · kg CO ₂ e	1.214941	0.891965 LOWER	kg CO ₂ e per FU	Craft amber ale, 440 ml tinplate steel can wins by 26.6%
Freshwater Eutrophication kg PO ₄ e	0.000193 LOWER	0.000261	kg PO ₄ e per FU	Craft amber ale, 330 ml glass bottle wins by 26.0%
Water Consumption Litres	17.102	14.499 LOWER	L per FU	Craft amber ale, 440 ml tinplate steel can wins by 15.2%

All values normalised to one functional unit. Lower is better for all categories. Neither product is strictly superior — the preferred choice depends on which environmental dimension matters most to the decision-maker.



Craft amber ale, 330 ml glass bottle

Edit product

Delete

1 litre of craft ale at point of consumer purchase

Scaling factor: 3.0303

A 330 ml bottle delivers 0.33 L per unit. All per-bottle impacts must be multiplied by $1000/330 \approx 3.0303$ to normalise to the 1 L functional unit. The glass bottle accounts for ~35% of the total GWP.

GWP

0.400931

Normalised: 1.214941
kg CO₂e · per unit / per FU

EUTROPHICATION

0.000064

Normalised: 0.000193
kg PO₄e · per unit / per FU

WATER CONSUMPTION

5.644

Normalised: 17102
L · per unit / per FU

Ingredient sourcing

Delete stage

MATERIAL	DESCRIPTION	QTY	UNIT	GWP KG CO ₂ E	EUTROPHICATION KG PO ₄ E	WATER L	
Malting barley, conventional	Base malt, pale ale malt (300 g/L recipe, per 330 ml batch)	0.1	kg	0.032	0.000028	0.21	Edit
Hops, dried, conventional	Dual-purpose hop variety, 5.5 g/L bitterness and aroma additions	0.0018	kg	0.00684	0.000002	0.063	Edit
Brewer's yeast, dry pitch	Dry ale yeast pitch, includes propagation losses	0.0006	kg	0.00099	0	0.007	Edit
Road freight transport, average truck	Malt and hops from supplier to brewery (avg 500 km, 0.104 kg total ingredients)	52.0	kg · km	0.004992	0.000003	0	Edit
Stage total				0.044822	0.000032	0.28	

+ Add flow to Ingredient sourcing

Brewing and fermentation

Delete stage

MATERIAL	DESCRIPTION	QTY	UNIT	GWP KG CO ₂ E	EUTROPHICATION KG PO ₄ E	WATER L	
Natural gas, combusted for heat EU	Wort boiling (60-minute boil, steam-jacketed kettle)	0.22	kWh	0.044	0.000001	0	Edit

ID	NAME	UNIT	REGION	GWP KG CO ₂ E / UNIT	EUTROPHICATION KG PO ₄ E / UNIT	WATER L / UNIT		
brewer_yeast	Brewer's yeast, dry pitch	kg	Global	1.65	0.00025	12.0	Edit	Delete
co2_beverage	Carbon dioxide, beverage grade	kg	Global	0.95	1.5e-05	0.3	Edit	Delete
diatomaceous_earth	Diatomaceous earth, filter aid	kg	Global	0.28	4e-05	0.2	Edit	Delete
electricity_grid_EU	Electricity, grid average	kWh	EU	0.25	2e-05	0.006	Edit	Delete
electricity_grid_GLO	Electricity, grid average	kWh	GLO	0.48	4.5e-05	0.01	Edit	Delete
electricity_grid_NL	Electricity, grid average	kWh	NL	0.31	2.5e-05	0.004	Edit	Delete
glass_primary	Glass, primary (container grade)	kg	Global	0.76	2.8e-05	4.5	Edit	Delete
dried_hops	Hops, dried, conventional	kg	Global	3.8	0.00085	35.0	Edit	Delete
malting_barley	Malting barley, conventional	kg	Global	0.32	0.00028	2.1	Edit	Delete
natural_gas_heat_EU	Natural gas, combusted for heat	kWh	EU	0.2	3e-06	0.001	Edit	Delete
natural_gas_heat_GLO	Natural gas, combusted for heat	kWh	GLO	0.22	5e-06	0.001	Edit	Delete
paper_and_board	Paper and paperboard	kg	Global	0.94	0.00028	0.1	Edit	Delete
phosphoric_acid	Phosphoric acid, 85%	kg	Global	1.18	0.00028	0.35	Edit	Delete
polypropylene	Polypropylene, general purpose	kg	Global	1.85	0.00062	0.08	Edit	Delete
process_water	Process water, industrial	L	Global	0.0003	5e-07	1.0	Edit	Delete

Requirements

- Parse and load the input data (both files should work)
- Product flow view showing details of one product
- Emission factors should be stored and editable
- Comparison view between two products with normalised comparison
- Manual product creation to create a new product
- (More details in the briefing file)



Non-Functional Requirements

- Application must run locally without requiring external API calls or cloud services
- All state must persist between sessions
- The comparison view must remain legible when impact scores differ by several orders of magnitude
- All generated source code must have approximately 80% test coverage



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WiFi: Studio N Guest
Password: werkspoor

Briefing & data:
<https://yoink.nl/hackathon.zip>